10TH CLASS COMPUTER SCIENCE — NEW SYLLABUS (PECTAA)

Chapter 8 — Digital Entrepreneurship

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Multiple Choice Questions

1. An example of intellectual property is:

(a) Shop furniture (b) Brand logo

(c) Raw materials (d) Office rent

Answer: (b) — Intellectual property is the creation of the mind of a person, such as an invention, a book, a song, an artwork, a brand name or a logo. A brand logo is the creative work of the business, so it is intellectual property. The furniture, the raw materials and the rent are physical things.

2. Which of the following laws protects personal data in the European Union?

(a) COPPA (b) GDPR

(c) DMCA (d) Cybercrime Act

Answer: (b) — GDPR stands for the General Data Protection Regulation, and it is the law which protects the personal data of the people in the European Union. COPPA is an American law which protects the data of the children under 13.

3. The document is required to register a trademark is:

(a) A bank statement (b) Proof of product purchase

(c) A copy of the logo and business name (d) Social media profile link

Answer: (c) — A trademark protects a brand name, a logo or a symbol. For its registration at the Intellectual Property Organization of Pakistan (IPO Pakistan), a copy of the logo and the business name is submitted, so that the mark is clearly identified.

4. Which of the following is a risk of poor data privacy?

(a) Increased customer trust (b) Cyberbullying

(c) Identity theft (d) Lower ad costs

Answer: (c) — When the personal data of the customers is not protected, it can be stolen by the hackers and misused, which leads to identity theft. In identity theft the criminal uses the name and the details of another person to commit a fraud.

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5. Which of the following is an example of personal data?

(a) Office address
(b) Employee ID card number
(c) Store opening hours
(d) Website theme color

Answer: (b) — Personal data is the information which belongs to a particular person, such as the name, the phone number, the address or the ID card number. The employee ID card number identifies one employee, so it is personal data; the office address and the opening hours belong to the business, not to a person.

6. A conflict of interest occurs when:

(a) Employees share company goals
(b) Personal benefit affects a business decision
(c) A business increases profits
(d) Customers give feedback

Answer: (b) — A conflict of interest occurs when the personal benefit or the relationship of a person can affect his business decision. For example, if a manager buys the material from his own relative's shop at a higher price, his personal interest has affected the decision of the business.

7. Which is part of digital ethics?

(a) Using false information in ads
(b) Protecting customer data respectfully
(c) Cyberbullying
(d) Avoid online presence

Answer: (b) — Digital ethics means using the internet and the digital tools responsibly and respectfully. Protecting the data of the customers respectfully is a clear part of it, while false advertising, cyberbullying and avoiding the online presence are not ethical business practices.

8. Which of the following is a CRM tool?

(a) Photoshop
(b) Zoho
(c) Google Maps
(d) Canva

Answer: (b) — CRM means Customer Relationship Management, that is, handling all the interactions with the customers to build trust. Zoho CRM is a well-known CRM tool which stores the names, the contact details and the purchase history of the customers in one system. Photoshop and Canva are design tools and Google Maps is a map service.

9. Which tool is used for marketing automation?

(a) Mailchimp
(b) PowerPoint
(c) MS Word
(d) Paint

Answer: (a) — Marketing automation means using special software to handle the marketing tasks automatically. Mailchimp automatically sends the welcome emails to the new customers, shares the special offers and posts the updates on the social media. PowerPoint, MS Word and Paint are not marketing tools.

10. In marketing, ROI means:

(a) Return on Investment (b) Rate of Information

(c) Revenue on Ideas (d) Reach of Internet

Answer: (a) — ROI stands for Return on Investment. It tells how much profit a business has received as compared to the money which it invested, and in this way it shows whether a marketing campaign was worth the money.

B. Short Questions

1. How does copyright protect creators in the digital world?

A copyright is the legal right which protects the original creative works of a person, such as books, songs, artwork, videos and software. In the digital world the work of a creator is very easy to copy and share, so the copyright protects it by giving the creator the exclusive right to reproduce, publish, sell or allow others to use his work. If anyone copies the work without permission, the creator can take legal action against him. In this way the copyright saves the income and the effort of the creator and encourages the people to create new works. In Pakistan it is registered at the Intellectual Property Organization (IPO Pakistan).

2. Differentiate between a patent and a trademark.

Patent	Trademark
It gives legal protection to a new product, design or method which a person has invented.	It protects a brand name, a logo or a symbol which identifies the goods or the services of a business.
It protects the **idea and the working** of the invention.	It protects the **identity and the name** of the business in the market.
After the approval, only the inventor can make, use or sell the invention for a certain period.	After the approval, no one else can use a similar name or logo for the same kind of business.
It is applied at the Patent Office under IPO Pakistan, with the details and the drawings of the invention.	It is applied at IPO Pakistan with a copy of the logo and the business name.

3. Define data privacy.

Data privacy means protecting the personal information of the people which the businesses collect, such as their names, addresses, phone numbers and payment details, and not sharing or using it without their permission. It means that the business collects only the information which it actually needs, uses it only for the purpose which was explained to the customer, and keeps it safe from the theft and the misuse. In Pakistan the Personal Data Protection Bill and in other countries the laws such as the GDPR and the COPPA make these rules compulsory.

4. Give one example of ethical decision-making in business.

A good example is the honest labelling of a product. If a shopkeeper finds that the expiry date of some of his products is near, he removes them from the shelf instead of selling them to the customers, even though this causes him a small loss. In this way he chooses what is fair, honest and responsible for the customer, and he saves the customers from a harm; this is an ethical business decision, and it also builds the trust of the customers in his shop.

5. Why is honesty important in business decisions?

Honesty means telling the truth and dealing fairly with everyone, and it is the base of a successful business. When the customers are given the correct information about the products and the prices, they trust the business and they come again and again, which brings a long-term profit. An honest business also gets the loyalty of its employees and the respect of its partners and the government. On the other hand, if a business uses false information or hides something, it may earn for a short time, but when the truth comes out, it loses its customers, its reputation and even its legal right to work.

6. Explain the purpose of data-driven marketing.

Data-driven marketing means making the marketing decisions on the basis of facts, numbers and analysis, instead of guesswork. Its purpose is to understand the customers correctly, so that the right product is offered to the right person at the right time. The business collects the data from the customers, the websites and the sales records and studies it to see which products are popular, which advertisement works and which group of people buys more. In this way the money of the advertising is spent on the useful campaigns only, the sales increase and the customers get the offers which are really useful for them.

7. What does SWOT analysis help identify?

SWOT analysis is a simple tool which helps the business to know its present situation. It identifies four things: **Strengths**, the things the business does well, such as high-quality products; **Weaknesses**, the areas which need improvement, such as slow customer service; **Opportunities**, the chances outside the business which it can use, such as a new market or a new technology; and **Threats**, the dangers and the competition which can harm it. With the help of this analysis the business plans its next steps wisely.

8. Explain the purpose of setting KPIs in marketing.

KPI stands for Key Performance Indicator. These are the measurable goals which tell whether a marketing campaign is working or not. Their purpose is to make the progress measurable and clear, instead of just hoping for the best. For example, a business which wants to sell its T-shirts within three months can set the KPIs such as the number of T-shirts sold every week, the number of the website visits and the number of the customer reviews. When the KPIs are compared with the actual results, the business comes to know which part of its marketing is successful and which part needs improvement.

C. Long Questions

1. Explain the process of problem solving and adaptability?

In the business world the challenges can appear at any time, such as losing the customers, facing the new competitors or a change in the technology. The successful entrepreneurs do not give up; instead they follow a clear method to understand the problem, to find its main cause and to take the effective steps.

Steps of problem solving:

(i) Identify the problem: first of all it is decided clearly what the problem actually is. For example, the sales of a shop have gone down.

(ii) Find the root cause: the reason behind the problem is searched by asking "why" again and again. For example, the customers are not coming back because of the poor packaging.

(iii) Think of the solutions: different possible solutions are written down, such as changing the packaging or the supplier.

(iv) Take action: the best solution is chosen and applied.

(v) Check the result: it is seen whether the problem has really been solved, and if needed, another step is taken.

For the planning, the **SWOT analysis** is used, which identifies the strengths, the weaknesses, the opportunities and the threats of the business, so that the business knows its situation and plans its next steps.

Adaptability and resilience: adaptability means being willing and able to change when it is needed, for example to adopt the new technology or to change the product according to the demands of the market. Resilience means that a business can recover quickly after a setback, such as a loss in one season. These two skills help the business to survive and to grow even in the difficult times. That is why every challenge is considered a hidden lesson: the businesses which stay curious, ask "why" often and act quickly turn their problems into the opportunities for growth.

2. Write a note on ethical business practices and digital ethics.

Ethical business practices: ethics means making the choices in business which are fair, honest and responsible. An ethical business treats its customers, employees and partners justly, keeps its promises and does not harm anyone for the sake of the profit. Its main points are the following.

(i) Honesty: the customer is given the correct information about the products, the prices and the quality, and no false promise is made.

(ii) Fairness: all the customers and the employees are treated equally, without any discrimination.

(iii) Avoiding a conflict of interest: a conflict of interest happens when the personal benefit or the relationship of a person can affect his business decision; an ethical business keeps the personal interest separate from the business decisions.

(iv) Responsibility: the business takes the responsibility of its mistakes and does not shift the blame to others.

Digital ethics: digital ethics means using the internet and the digital tools responsibly and respectfully. It includes the responsible marketing, in which the products and the services are promoted honestly, the customer is given correct and complete information and the false promises are avoided. It also includes the protection of the data of the customers, the respect of their privacy, and the keeping of a respectful environment on the social media, without any cyberbullying. The digital businesses must also follow the data privacy laws, such as the Personal Data Protection Bill of Pakistan, the GDPR of the European Union and the COPPA of the United States.

In short, the ethical practices build the trust of the customers, and the trust is the most valuable asset of any business in the digital age.

3. (a) Discuss how CRM tools and loyalty programmes help in retaining customers. (b) Explain Marketing automation and describe their benefit for small businesses.

(a) CRM tools and loyalty programmes:

CRM means Customer Relationship Management, that is, handling all the interactions with the customers in one place to build trust and long-term relationships. A CRM tool stores the names, the contact details and the purchase history of the customers in one system. For example, a sports shop in Islamabad uses Zoho CRM to store the names, the contact details, the shoe sizes and the purchase history of its customers. When a new sports shoe arrives in the size which a particular customer usually buys, the shop sends him a quick message or an email about it. The customer feels that the shop knows his needs, so he comes back to the same shop.

A loyalty programme rewards the customers for buying again and again. For example, a coffee shop in Karachi gives a free drink after every ten purchases, and it also sends personalized emails with the name of the customer and the offers of his favourite drink. Such rewards and the personal attention make the customers feel valued, so they return to the business and do not go to the competitor. In this way the CRM tools and the loyalty programmes together increase the sales and reduce the cost, because it is much cheaper to keep an old customer than to find a new one.

(b) Marketing automation:

Marketing automation means using special software to handle the marketing tasks automatically, without doing each step by hand. For example, **Mailchimp** can automatically send a welcome email to every new customer, share the special offers and post the updates on the social media. In the same way, the reminders, the birthday wishes and the follow-up emails are sent by the software on the set time.

Benefits for small businesses:

(i) Saves time and cost: the small business does not need a big marketing team, because the software does the regular work by itself.

(ii) Works all the time: the emails and the posts are sent even at night and on the holidays, so the business stays connected with the customers.

(iii) Reaches more customers: a large number of customers are contacted at the same time, which expands the customer reach of a small business.

(iv) Personal and measurable: the messages can be personalized with the name of the customer, and the results such as the open rate and the sales can be measured, so the next campaign becomes better. In this way the marketing automation improves the efficiency, reduces the operational cost and helps the small business to grow.

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